

Normalization

Need to “normalize” terms

Information Retrieval: indexed text & query terms must have same form.

We want to match **U.S.A.** and **USA**

We implicitly define equivalence classes of terms

e.g., deleting periods in a term

Alternative: asymmetric expansion:

Enter: **window** Search: **window, windows**

Enter: **windows** Search: **Windows, windows, window**

Enter: **Windows** Search: **Windows**

Potentially more powerful, but less efficient

Morphology

Morphology is the study of the internal structure of words, of the way words are built up from smaller meaning units.

Morpheme:

The smallest meaningful unit in the grammar of a language.

Two classes of morphemes

Stems: "main" morpheme of the word, supplying the main meaning (i.e. *establish* in the example below)

Affixes: add additional meaning

Prefixes: **Antidisestablishmentarianism**

Suffixes: **Antidisestablishmentarianism**

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Morphology

Morphemes:

The small meaningful units that make up words

Stems: The core meaning-bearing units

Affixes: Bits and pieces that adhere to stems
Often with grammatical functions

Types of morphological processes

Inflection:

Systematic modification of a root form by means of prefixes and suffixes to indicate grammatical distinctions like singular and plural.

Stems: also called lemma, base form, root, lexeme

Doesn't change the word class

New grammatical role

Usually produces a predictable, non idiosyncratic change of meaning.

run → runs | running | ran

hope+ing → hoping hop → hopping

Derivation:

Ex: *compute* → *computer* → *computerization*

Less systematic than inflection

It can involve a change of meaning

Wide → *Widely*

Suffix *en* transforms adjective into verbs

Weak → *weaken*, *soft* → *soften*

Suffix *able* transforms verbs into adjective

Understand → *Understandable*

Suffix *er* transforms verbs into nouns (nominalization)

teach → *teacher*

NLTK Stemmers

NLTK includes several off-the-shelf stemmers. The Porter and Lancaster stemmers follow their own rules for stripping affixes.

Stemming is not a well-defined process, and we typically pick the stemmer that best suits the application we have in mind.

```
>>> raw = "DENNIS: Listen, strange women lying in ponds distributing swords  
... is no basis for a system of government. Supreme executive power derives from  
... a mandate from the masses, not from some farcical aquatic ceremony."  
>>> tokens = nltk.word_tokenize(raw)
```

```
>>> porter = nltk.PorterStemmer()  
>>> lancaster = nltk.LancasterStemmer()  
>>> [porter.stem(t) for t in tokens]  
['DENNI', ':', 'Listen', ',', 'strang', 'women', 'lie', 'in', 'pond',  
'distribut', 'sword', 'is', 'no', 'basi', 'for', 'a', 'system', 'of', 'govern',  
, '.', 'Suprem', 'execut', 'power', 'deriv', 'from', 'a', 'mandat', 'from',  
'the', 'mass', ',', 'not', 'from', 'some', 'farcic', 'aquat', 'ceremoni', '.']  
>>> [lancaster.stem(t) for t in tokens]  
['den', ':', 'list', ',', 'strange', 'wom', 'lying', 'in', 'pond', 'distribut',  
'sword', 'is', 'no', 'bas', 'for', 'a', 'system', 'of', 'govern', '.', 'suprem',  
'execut', 'pow', 'der', 'from', 'a', 'mand', 'from', 'the', 'mass', ',', 'not',  
'from', 'som', 'farc', 'aqu', 'ceremony', '.']
```

Preprocessing: Stemming

Stemming & Lemmatization = Cut word down to base form

- Stemming: Uses rough heuristics to reduce words to base
- Lemmatization: Uses vocabulary and morphological analysis
- Makes the meaning of run, runs, running, ran all the same
- Cuts down on complexity by reducing the number of unique words

Multiple stemmers available in NLTK

- PorterStemmer, LancasterStemmer, SnowballStemmer
- WordNetLemmatizer

N-gram

- An n-gram is a sequence of n tokens from a given sample of text or speech.
- We can include n-grams in our term frequencies too.
- They are basically a set of co-occurring words within a given window and when computing the n-grams you typically move one word forward (although you can move X words forward in more advanced scenarios).

Sentence	1-gram (uni-gram):	2-gram (bi-gram):
It is not a dog, it is a wolf	"it", "is", "not", "a", "dog", "it", "is", "a", "wolf"	"it is", "is not", "not a", "a dog", "dog it", "it is", "is a", "a wolf"

For example, for the sentence "The cow jumps over the moon".

If $N=3$, the n-grams would be:

1. the cow jumps
2. cow jumps over
3. jumps over the
4. over the moon

So you have 4 n-grams in this case. When $N=1$, this is referred to as **unigrams** and this is essentially the individual words in a sentence. When $N=2$, this is called **bigrams** and when $N=3$ this is called **trigrams**. When $N>3$ this is usually referred to as four grams or five grams and so on.

How many N-grams in a sentence?

If $X = \text{Num of words in a given sentence } K$, the number of n-grams for sentence K would be:

$$Ngrams_K = X - (N - 1)$$